

Sina Sajadmanesh

📍 Zurich, Switzerland ✉ sina.sajadmanesh@gmail.com 🌐 sajadmanesh.com 📷 sajadmanesh

🔗 sisaman

Education

Swiss Federal Institute of Technology (EPFL), PhD in Electrical Engineering May 2019 – Aug 2023
Sharif University of Technology, MSc in Information Technology Engineering Sept 2014 – Sept 2016
University of Isfahan, BSc in Computer Software Engineering Sept 2009 – Feb 2014

Experience

AI Engineer, Sony AI – Zurich, Switzerland Oct 2023 – present

- Working on privacy-preserving vision foundation models
- Implemented easy-to-use APIs for vision foundation model inference and deployment
- Developed a comprehensive multi-task learning framework based on OpenMMLab libraries and PyTorch, supporting various vision tasks, new model architectures, and mixed-precision and distributed training
- Implemented an end-to-end open-world classification pipeline using MMClassification, HuggingFace transformers, and CLIP
- Improved model training and inference speed by optimizing the data pipeline, using mixed-precision training, and leveraging distributed training on multiple GPUs

Research Assistant, Idiap Research Institute – Martigny, Switzerland May 2019 – Aug 2023

- Worked as a doctoral researcher on differentially private machine learning with graph neural networks
- **3 published papers** in top-tier conferences (CCS, USENIX Security, and NDSS) with **185+ citations**
- **7 invited talks** at top universities and research institutions, including Imperial College, UIC, and Twitter
- **6 open-source projects** with **110+ stars** on GitHub
- **1 short course** taught on "Trustworthy Machine Learning" at [Artificial Intelligence Doctoral Academy](#)
- **Finalist** in [CSAW Applied Research Competition](#) for the best paper award in computer security in Europe

Visiting Collaborator, The Alan Turing Institute – London, UK Mar 2023 – Mar 2023

- Co-organized a workshop on "[Privacy and Fairness in AI for Health](#)" with 60+ attendees

Research Intern, Brave Software – Remote Mar 2022 – May 2022

- Developed a novel privacy-preserving **federated learning** framework for **neural bandit models** under client heterogeneity using PyTorch, Flower, Dask, and Tensorflow-Lite

Research Assistant, Sharif University of Technology – Tehran, Iran Nov 2014 – May 2019

- Worked on various research projects including privacy-preserving machine learning, web data science, and social and information network analysis.
- **4 published papers** in top-tier venues (WWW, TKDD, IoTJ, ASONAM) with **430+ citations**
- **5+ press releases** in top-tier media outlets, including [MIT Technology Review](#), [France 24](#), and [The Independent](#)
- **2 open-source projects** with **20+ stars** on GitHub
- **1 semester course** taught on "Fundamentals of Programming with Python" at Sharif University of Technology

Skills and Expertise

Programming Languages: Python, C++, Java, SQL, Shell, \LaTeX

Machine Learning and AI: PyTorch, TensorFlow, HuggingFace, OpenMMLab, PyTorch-Lightning

MLOps and DevOps: Weights & Biases, Docker, GitHub, Dask, Neptune, Linux

Privacy-Enhancing Technologies: Flower, Opacus, Auto-DP

Community and Professional Service

Invited Speaker: Imperial College London (2023, 2020), University of Illinois at Chicago (2022), L3S Research Center (2022), Graph Neural Networks User Group Meetup (2021), Twitter Machine Learning Seminar (2021)

Organizing Committee: [Privacy and Fairness in AI for Health](#) (2023)

Program Committee: [PPAI](#) (2024), [WiseML](#) (2023), [PAIR2Struct](#) (2022), [DPML](#) (2021)

Reviewer: [NeurIPS](#) (2024), [IMWUT](#) (2024), [TDSC](#) (2023), [LoG](#) (2023, 2022), [AISTATS](#) (2023), [AIJ](#) (2022), [TBD](#) (2021), [TIST](#) (2020), [SNAM](#) (2020), [WWWJ](#) (2018)

Publications

Argus: A Compact and Versatile Foundation Model for Vision June 2025

Weiming Zhuang, Chen Chen, Zhizhong Li, Sina Sajadmanesh, et al.
 The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

ProGAP: Progressive Graph Neural Networks with Differential Privacy Mar 2024

Guarantees

Sina Sajadmanesh, Daniel Gatica-Perez
 ACM International Conference on Web Search and Data Mining (WSDM)

Privacy-Preserving Machine Learning on Graphs Aug 2023

Sina Sajadmanesh

Doctoral Thesis - Swiss Federal Institute of Technology (EPFL)	
<u>GAP: Differentially Private Graph Neural Networks with Aggregation Perturbation</u>	Aug 2023
Sina Sajadmanesh, Ali Shahin Shamsabadi, Aurélien Bellet, et al. USENIX Security Symposium (USENIX Security)	
<u>Locally Private Graph Neural Networks</u>	Nov 2021
Sina Sajadmanesh, Daniel Gatica-Perez ACM Conference on Computer and Communications Security (CCS)	
<u>When Differential Privacy Meets Graph Neural Networks</u>	June 2020
Sina Sajadmanesh, Daniel Gatica-Perez Technical Report, ArXiv e-prints	
<u>A Hybrid Deep Learning Architecture for Privacy-Preserving Mobile Analytics</u>	May 2020
Seyed Ali Ossia, Ali Shahin Shamsabadi, Sina Sajadmanesh, et al. IEEE Internet of Things Journal (IoTJ)	
<u>Continuous-Time Relationship Prediction in Dynamic Heterogeneous Information Networks</u>	Aug 2019
Sina Sajadmanesh, Sogol Bazargani, Jiawei Zhang, Hamid R. Rabiee ACM Transactions on Knowledge Discovery from Data (TKDD)	
<u>NPGLM: A Non-Parametric Method for Temporal Link Prediction</u>	June 2017
Sina Sajadmanesh, Jiawei Zhang, Hamid R. Rabiee Technical Report, ArXiv e-prints	
<u>Kissing Cuisines: Exploring Worldwide Culinary Habits on the Web</u>	Apr 2017
Sina Sajadmanesh, Sina Jafarzadeh, Seyed Ali Ossia, et al. International World Wide Web Conference Companion (WWW)	
<u>Predicting Anchor Links between Heterogeneous Social Networks</u>	Aug 2016
Sina Sajadmanesh, Hamid R. Rabiee, Ali Khodadadi International Conference on Advances in Social Networks Analysis and Mining (ASONAM)	